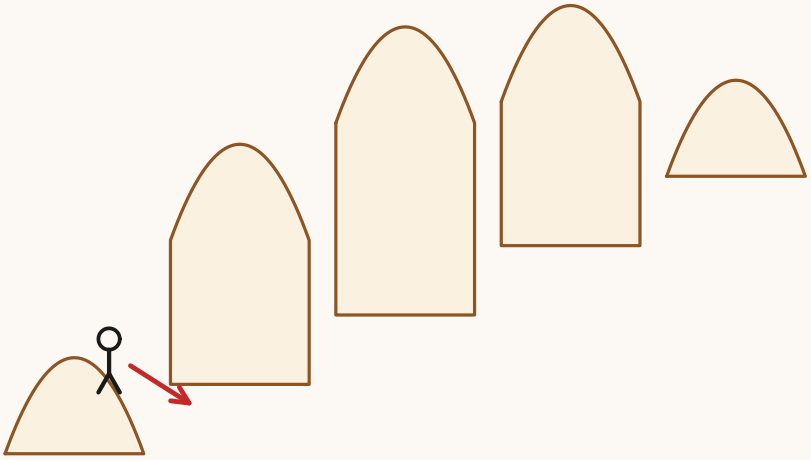
Chapter 38

Partial differentiation

one direction at a time



a hill is a height that depends on two things: how far east and how far north you are

Every function so far had one input: put in x, get out y. Plenty of real things depend on **two** inputs. The height of a hill depends on how far east and how far north you stand; the volume of a can depends on its radius and its height.

You can't ask "what's the slope?" of a hill: it depends which way you face. So we ask one direction at a time, keeping everything else still. That is **partial differentiation**.

Goes with **Lecture 14** (the notation slide: "∂y/∂x is used when the function is made up of more than one variable"). Not a lecture of its own, but it builds straight on Chapter 34.

Two inputs, one output: a height table

Take the hill $h = 12 - x^2 - 2y^2$, where x is metres east and y metres north of the summit. Every pair (x, y) gives one height, so a spreadsheet of heights describes the hill. Darker means higher:

	x = -2	x = -1	x = 0	x = 1	x = 2
y = 2	0	3	4	3	0
y = 1	6	9	10	9	6
y = 0	8	11	12	11	8
y = -1	6	9	10	9	6
y = -2	0	3	4	3	0

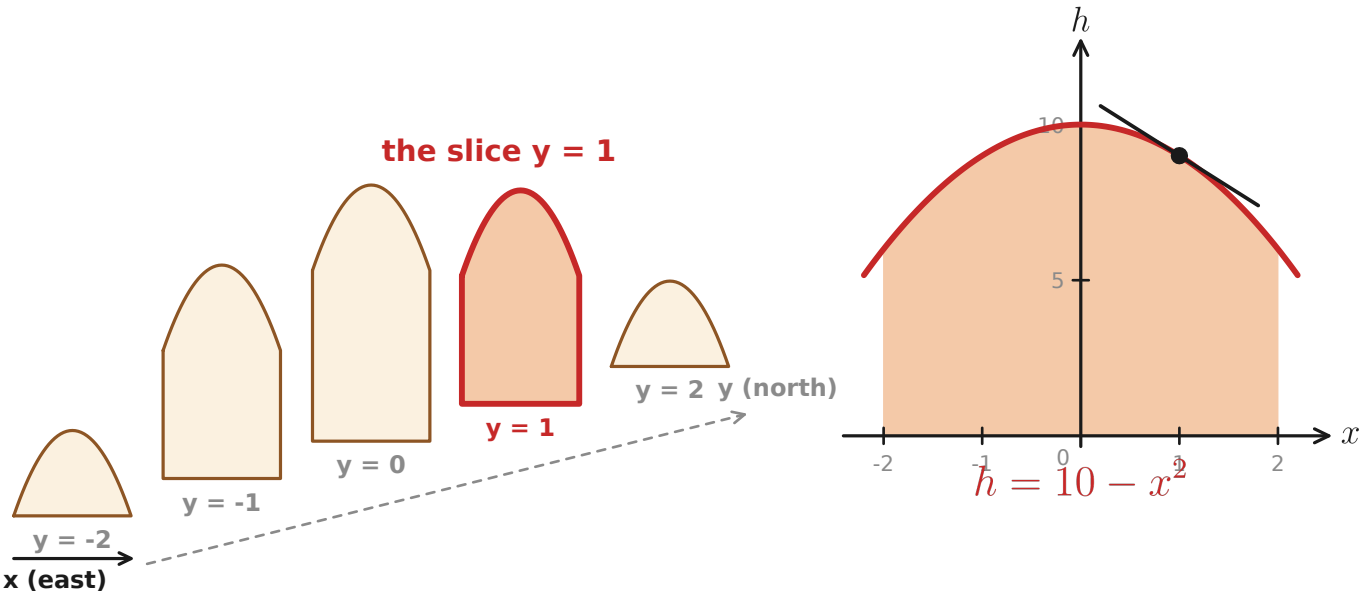
the red cell: 1 m east, 1 m north.
h = 12 − 1 − 2 = 9

walk along a row (east): the heights go 9, 10, 9, 6 … gently.
walk up a column (north): 3, 9, 11, 9, 3: steeper.

We write $h(x, y)$: "h of x and y". The question for the rest of the chapter: **how steep is it** at the red cell? Answer: it depends which way you walk.

Slice the loaf

Draw the hill in 3D and it looks like a loaf of bread. Cut it into slices, each at a fixed y. **Every slice is an ordinary curve**, with only x changing, and Chapter 34 already knows how to find the slope of those.



With y stuck at 1 the formula becomes $h = 12 - x^2 - 2 = 10 - x^2$, a plain one-letter curve. Its slope is $-2x$, so at $x = 1$ it's -2: walking east there, you drop 2 m for every metre.

Freeze the other letter

RULE: partial differentiation

$\frac{\partial f}{\partial x}$: differentiate with respect to x while **holding every other letter constant**. Treat y exactly like a number such as 7.

So a term with no x in it (like $3y^2$) differentiates to **0**, and a frozen letter multiplying x stays as a multiplier: $\frac{\partial}{\partial x}(yx^2) = 2xy$.

All the Chapter 34–35 rules (power, chain, product, quotient) still apply, to the moving letter only.

Subscript notation: f_x means $\frac{\partial f}{\partial x}$ and f_y means $\frac{\partial f}{\partial y}$.

Slicing at $y = 1$ worked, but we'd like a formula for **any** slice. So don't put in a number: just **pretend** y is a number. Picture it frozen in an ice cube. It can't move, so it behaves exactly like the 7 in $7x^2$:

$7x^2 \longrightarrow 14x$

a number in front stays in front

$y\,x^2 \longrightarrow 2x\,y$

so does a frozen y

$7 \longrightarrow 0$

a number alone vanishes

$y^2 \longrightarrow 0$

so does a frozen y on its own

The new notation: a curly ∂ (say "partial" or "del") instead of d, so nobody forgets that another letter was frozen:

$$\frac{\partial h}{\partial x}$$

"partial h by partial x": slope walking east, y frozen

The hill, both ways. $h = 12 - x^2 - 2y^2$.

$$\frac{\partial h}{\partial x} = 0 - 2x - \underbrace{2\,y^2}_{\text{frozen} \to 0} = -2x$$

← 12 → 0 as usual

$$\frac{\partial h}{\partial y} = 0 - \underbrace{x^2}_{\text{frozen} \to 0} - 4y = -4y$$

← now freeze x instead

$$\text{at } (1,1): \quad \frac{\partial h}{\partial x} = -2, \qquad \frac{\partial h}{\partial y} = -4$$

← north is twice as steep

That matches the height table: the column (north) changed faster than the row (east).

Term by term, ice cube by ice cube

Example. $f = x^2y + 3y^2 + 5x$. Go through one term at a time, asking "does this term contain the moving letter?"

term	∂/∂x (freeze y)	∂/∂y (freeze x)
x^2y	$2x\ y$	x^2
$3y^2$	0	$6y$
$5x$	5	0

$$\frac{\partial f}{\partial x} = 2xy + 5, \qquad \frac{\partial f}{\partial y} = x^2 + 6y$$ ← add the columns

Lecture 14's example, printed as $y = 2x^2x + 3z$. Here y is the output and the inputs are x and z. As printed, $2x^2x = 2x^3$:

$$\frac{\partial y}{\partial x} = 6x^2, \qquad \frac{\partial y}{\partial z} = 3$$

If it was meant to be $2x^2z + 3z$ (a more typical example), freeze z for the first and x for the second:

$$\frac{\partial y}{\partial x} = 4x\,z, \qquad \frac{\partial y}{\partial z} = 2x^2 + 3$$

With Chapter 35's rules. A frozen letter changes nothing about the chain rule. It's just a number riding along:

$$z = e^{2x}\sin y: \quad \frac{\partial z}{\partial x} = 2e^{2x}\sin y, \quad \frac{\partial z}{\partial y} = e^{2x}\cos y$$

sin y or e^{2x} is a constant factor

$$z = \ln(x^2 + y^2): \quad \frac{\partial z}{\partial x} = \frac{2x}{x^2 + y^2}$$ ← chain rule: inside's slope ÷ inside

RULE: when the product rule is needed

Look at the factors of a product. If only **one** factor contains the moving letter, the other is a constant multiplier: no product rule.

If **both** factors contain the moving letter, use the product rule $vu' + uv'$, with $'$ meaning $\frac{\partial}{\partial x}$ (or $\frac{\partial}{\partial y}$).

Example. $z = xe^{xy}$. Both factors, x and e^{xy}, contain x:

$$u = x, \quad v = e^{xy}, \quad \frac{\partial u}{\partial x} = 1, \quad \frac{\partial v}{\partial x} = y\,e^{xy} \longleftarrow \text{chain: inside } xy \to y$$

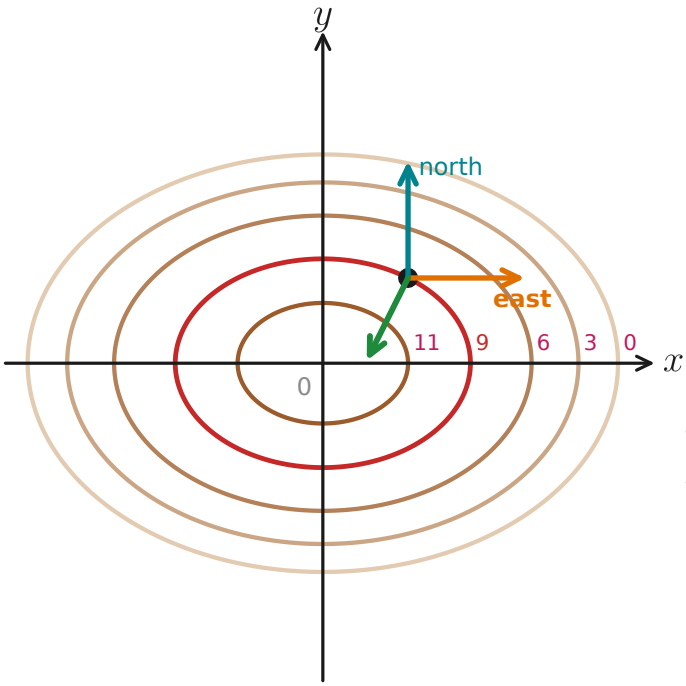
$$\frac{\partial z}{\partial x} = e^{xy} \cdot 1 + x \cdot ye^{xy} = (1 + xy)\,e^{xy} \longleftarrow \text{product rule}$$

Now with respect to y: the first factor x is frozen, so it is just a multiplier:

$$\frac{\partial z}{\partial y} = x \cdot x\,e^{xy} = x^2 e^{xy} \longleftarrow \text{no product rule; chain: } xy \to x$$

The walker's map: contour lines

Hikers don't carry 3D models. They carry a **contour map**: lines joining points of equal height, seen from above. For our hill the contours are ovals. Stand at $(1, 1)$, height 9:



orange: walk east. Slope ∂h/∂x = −2.
teal: walk north. Slope ∂h/∂y = −4 (the contours are more tightly packed that way: closer lines = steeper).

the red oval is height 9, the one you're standing on (12 − 1² − 2 × 1² = 9). Walk along it and the height doesn't change at all.

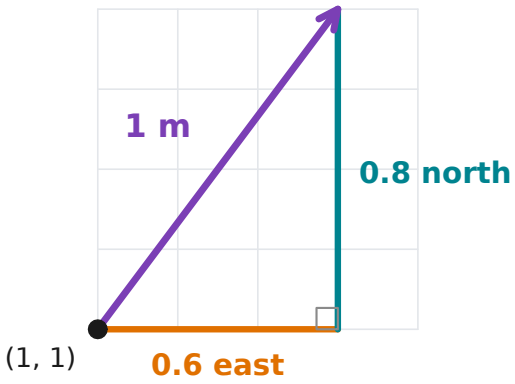
The uphill arrow. Put the two partial slopes together as a vector (Chapter 32):

$$\nabla h = \begin{pmatrix} \partial h/\partial x \\ \partial h/\partial y \end{pmatrix} = \begin{pmatrix} -2 \\ -4 \end{pmatrix} \text{ at } (1, 1)$$ ← *say "grad h"*

This arrow (green, scaled down, on the map) points the **steepest way uphill** and cuts across the contour at a right angle. It points south-west, roughly at the summit, but not exactly: it leans **south** (its y-part, -4, is bigger than its x-part, -2), because the hill is steeper in the north–south direction. Its length, $\sqrt{(-2)^2 + (-4)^2} = \sqrt{20} \approx 4.47$, is how steep that steepest way is.

Walking any other way. Most walks aren't due east or due north. Walk diagonally, and each metre you go splits into an **east part** and a north part:

> ✎ **YOUR GO (pencil first)**
>
> *From (1, 1) you walk 1 m in a direction that takes you 0.6 m east and 0.8 m north. East drops you 2 per metre, north drops you 4 per metre. Roughly how much height do you lose?*



east part: 0.6 × (−2) = −1.2
north part: 0.8 × (−4) = −3.2

total: about 4.4 lost per metre of diagonal walking.

That's the **directional slope**: take each partial slope, weight it by how much of your step goes that way, and add. It's the dot product (Chapter 32) of the uphill arrow with your walking direction: $(-2)(0.6) + (-4)(0.8) = -4.4$. Walk straight along ∇h and you get the steepest possible climb, 4.47; walk along the contour and you get 0.

Second partials: the mixed ones agree

RULE: second partial derivatives and their notation

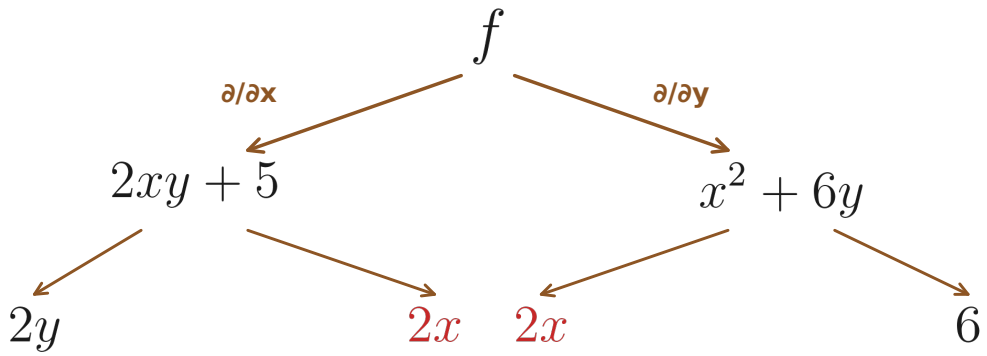
$\frac{\partial^2 f}{\partial x^2} = f_{xx}$: differentiate by x twice. $\frac{\partial^2 f}{\partial y^2} = f_{yy}$: by y twice.

$\frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)$: differentiate by x **first**, then by y. Read the bottom **right to left**.

In subscripts the order reads **left to right**: $f_{xy} = (f_x)_y$, x first then y. So $f_{xy} = \frac{\partial^2 f}{\partial y\,\partial x}$.

For the smooth functions in this course the two mixed partials are equal, so the order never changes the answer; it is still worth reading the notation correctly.

You can differentiate again, in either letter. With $f = x^2y + 3y^2 + 5x$ from before:

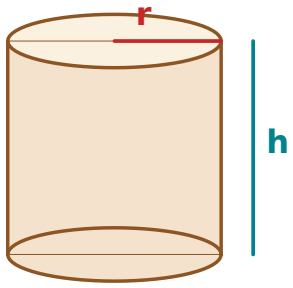


$$\frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial^2 f}{\partial x\,\partial y}$$ ← mixed partials are equal

the two middle routes (x then y, y then x) land on the same 2x. They always do for the smooth functions you'll meet.

What partials are for: small changes

A can's volume is its circle's area times its height: $V = \pi r^2 h$. A factory makes cans with $r = 2$ cm and $h = 5$ cm, but the machine drifts: r grows by 0.1 and h shrinks by 0.2. Roughly how much does the volume change?



Each partial says how sensitive V is to one measurement while the other stays put. Multiply each by its own nudge and add (the δ's are Lecture 14's "small increments"):

$$\delta V \approx \frac{\partial V}{\partial r}\,\delta r + \frac{\partial V}{\partial h}\,\delta h$$

$$\frac{\partial V}{\partial r} = 2\pi r h = 20\pi, \qquad \frac{\partial V}{\partial h} = \pi r^2 = 4\pi \quad \longleftarrow \text{at } r = 2,\ h = 5$$

$$\delta V \approx 20\pi(0.1) + 4\pi(-0.2) = 2\pi - 0.8\pi = 1.2\pi \approx 3.77 \text{ cm}^3$$

Check with the real volumes: $\pi(2.1^2)(4.8) - \pi(2^2)(5) \approx 3.67$. Close, and the smaller the nudges, the closer it gets.

Find the unknown

PROCEDURE: a partial derivative with a missing value

1. Differentiate with respect to the stated letter, treating the unknown constant (a, k...) and the other letters as numbers.
2. Substitute the given point (x, y).
3. Set the result equal to the given value and solve for the unknown. Check by substituting back.

Example. $f = kxy^2 + x^3$ has $\frac{\partial f}{\partial y} = 8$ at $(1, 2)$. Find k.

$$\frac{\partial f}{\partial y} = 2kxy$$ ← x and k frozen; $x^3 \to 0$

$$2k(1)(2) = 8 \;\Rightarrow\; 4k = 8 \;\Rightarrow\; k = 2$$ ← at (1, 2)

Example. Where are both slopes of $z = x^2 + y^2 - 4x + 6y$ zero (the bottom of the bowl)?

$$\frac{\partial z}{\partial x} = 2x - 4 = 0 \Rightarrow x = 2, \qquad \frac{\partial z}{\partial y} = 2y + 6 = 0 \Rightarrow y = -3$$

Both are zero at $(2, -3)$: the surface is flat in every direction there.

✗ Common mistakes

✗ $\frac{\partial}{\partial x}(3y^2) = 6y$

no x in the term: it is a constant

✓ $\frac{\partial}{\partial x}(3y^2) = 0$

✗ $\frac{\partial}{\partial x}(x^2 y) = 2x + x^2$

frozen y is a multiplier, not a product

✓ $\frac{\partial}{\partial x}(x^2 y) = 2xy$

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Partial derivative	$\dfrac{\partial f}{\partial x} = f_x$ differentiate by x, holding every other letter constant
Frozen letters	a term with no moving letter → 0; a frozen factor stays as a multiplier
Product rule	only when both factors contain the moving letter
Second partials	$f_{xx} = \dfrac{\partial^2 f}{\partial x^2}, \qquad f_{xy} = \dfrac{\partial^2 f}{\partial y\,\partial x}$ f_{xy}: x first, then y
Mixed partials	$\dfrac{\partial^2 f}{\partial y\,\partial x} = \dfrac{\partial^2 f}{\partial x\,\partial y}$ for smooth functions
Gradient vector	$\nabla f = \begin{pmatrix} f_x \\ f_y \end{pmatrix}$ points steepest uphill
Small changes	$\delta z \approx \dfrac{\partial z}{\partial x}\delta x + \dfrac{\partial z}{\partial y}\delta y$
Find the unknown	differentiate, substitute the point, set equal to the given value, solve

Your turn

Freeze, then differentiate. Answers in grey.

Freeze the other letter

1. $f = x^3 + 4xy + y^2$: find $\partial f/\partial x$ and $\partial f/\partial y$
2. $f = 5x^2y^3$: find $\partial f/\partial x$ and $\partial f/\partial y$
3. $z = 7y - 3x$ (careful: very short answers)

1. $3x^2 + 4y$, $4x + 2y$ 2. $10xy^3$, $15x^2y^2$ 3. $\partial z/\partial x = -3$, $\partial z/\partial y = 7$

With the chain rule

4. $z = \sin(3x + 2y)$
5. $z = e^{(xy)}$

4. $3\cos(3x + 2y)$, $2\cos(3x + 2y)$ 5. $y\, e^{(xy)}$, $x\, e^{(xy)}$

The hill $h = 12 - x^2 - 2y^2$

6. Find both slopes at $(2, -1)$.
7. Which way is steeper there, east or north?

6. $\partial h/\partial x = -4$, $\partial h/\partial y = 4$ 7. equally steep (both size 4); east goes down, north goes up

Second partials

8. $f = x^2y^3$: check that the two mixed partials agree.

8. both are $6xy^2$

Small changes

9. A rectangle is 10 by 5. Each side grows by 0.1. Estimate the change in area $A = xy$.

9. $\delta A \approx 5(0.1) + 10(0.1) = 1.5$ (true: 1.51)

Product rule and notation

10. $z = x\, e^{(xy)}$: find $\partial z/\partial x$ and $\partial z/\partial y$.
11. $z = x^2 \sin(xy)$: find $\partial z/\partial x$.
12. $f = x^3y^2$: find f_x, f_y and f_{xy}.

10. $(1 + xy)e^{(xy)}$, $x^2e^{(xy)}$ 11. $2x \sin(xy) + x^2y \cos(xy)$ 12. $3x^2y^2$, $2x^3y$, $6x^2y$

Find the unknown

13. $f = ax^2y + 3y^2$ and $\partial f/\partial x = 12$ at $(1, 2)$. Find a.
14. $f = x^3 + bxy$ and $\partial f/\partial x = 0$ at $(1, 3)$. Find b.
15. f = e^(kxy) and $\partial f/\partial x = 6$ at $(0, 3)$. Find k.
16. $z = x^2 + y^2 - 6x + 2y$: find the point where both partials are zero.
17. $V = \pi r^2 h$ with $r = 2$. Find h so that $\partial V/\partial r = \partial V/\partial h$.

13. $a = 3$ ($2axy = 4a = 12$) 14. $b = -1$ ($3 + 3b = 0$) 15. $k = 2$ ($ky\,e^0 = 3k = 6$) 16. $(3, -1)$ ($2x - 6 = 0$, $2y + 2 = 0$) 17. $h = 1$ ($2\pi rh = \pi r^2$, so $4\pi h = 4\pi$)